AA273 Final Project Proposal: LED-Based Beacon System for Relative Positioning and Pose Estimation

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***Abstract*—In this paper we introduce a novel LED-based beacon system to obtain relative position and orientation estimates for two formation-flying satellites with stringent relative positioning and attitude requirements. The LED beacons are attached to the face of one satellite (Deputy) in a known arrangement, and a camera on the other satellite (Chief) continuously images this beacon and converts the image data into relative distance and orientation information. The true state of the deputy satellite will be estimated using a Kalman Filter (KF) that uses the beacon system's estimates at every time-step as its inputs. This beacon system is designed to surpass the accuracy constraints of traditional GPS-based positioning systems to deliver fine positioning and attitude knowledge. Our results showed millimeter-level accuracy for data corrupted with noise characteristic of CCD sensors in cameras, and centimeter-scale accuracy for data corrupted by shot noise from starlight.**

***Index Terms*—Relative Positioning, Pose Estimation, LED beacons, Computer vision**

I. Introduction and Literature Review

Satellite formations and swarms often impose stringent relative positioning and attitude requirements that are important from both a swarm-safety perspective (i.e. keeping a minimum separation distance between satellites for collision avoidance) as well as a mission objective perspective. As satellite swarms increase in the number of elements, inter-satellite communication and ranging operations will increase in complexity, necessitating distributed or decentralized architectures over centralized architectures. Such architectures require a low-SWaPC method of inter-satellite ranging and pose estimation that works in real-time and is computationally easy to implement.

From a mission objective perspective, some swarms have stringent requirements on inter-satellite positioning and accuracy. The upcoming STARI mission [1] is a planned technology demonstration mission to demonstrate the reflection of starlight from one CubeSat to another in space for the first time, thus enabling space-based interferometry using formation-flying satellites for astronomical applications. This mission requires navigational accuracy on the sub-centimeter scale between two CubeSats separated on the 100 m-scale, while also requiring pointing accuracy on the 10 arcsecond-scale and relative positioning on the mm-scale, all in real-time to maintain reflection over a designated time period (on the order of 10 seconds). The VISORS mission [2], designed to image the sun through two formation-flying spacecraft, has similarly stringent positioning and attitude requirements, requiring 90-arcsecond pointing precision at a 40-m separation with mm-scale relative positioning requirements. Satellite missions aiming to perform Rendezvous, Proximity Operations, and Docking (RPOD) also have similarly stringent positioning requirements that usually scale with the size of the spacecraft, with CubeSat-based RPOD missions requiring sub-cm accuracy for docking [3].

GPS is inadequate for such requirements, with even state-of-the-art methods such as Carrier Phase Differential GPS failing to achieve sub-cm accuracy in real time, instead achieving real-time accuracy only on the order of 2 cm [4].

The concept of using LEDs to track satellites is not new, but much of the literature on the subject involves ground-based tracking of objects in space. [5] successfully demonstrated tracking of a balloon with LEDs using a ground-based telescope. [6] extended this concept to track a 1U CubeSat with LEDs in Low-Earth Orbit, with several successful observations [7]. [8] has attempted to standardize a "license-plate" for satellites using LEDs flashing with distinct frequencies that can be tracked from the ground. However, the concept of using LEDs for space-based tracking has been sparsely explored. [9] developed a relative navigation package for CubeSats that uses LEDs on a single panel to deduce relative positioning and orientation information, and their lab tests produced results with positional accuracy of 5 mm and attitude accuracy of 2°. This paper used centroiding to locate the LEDs and a simplified pin-hole camera model to find the resultant rotation matrix in their tests, which is similar to the methodology followed in this proposal (as described in following sections). [10] also proposed the use of LEDs for relative navigation of CubeSats, placing LEDs on all six sides of the CubeSat and conducting their tests in a facility akin to that used for motion-capture. They also chose to employ a Convolutional Neural Network for their LED acquisition algorithm, with mixed results due to smaller training and validation datasets.

Altogether, work on relative positioning and pose estimation in space for satellite formations or swarms remains sparse and untested. The purpose of this project is to bridge the gap in literature and engineering by demonstrating the efficacy

of a CubeSat-based LED system for relative navigation in the context of the STARI mission. Successful simulation and filtering for position and pose estimation through AA 273 will constitute the first phase of this project and pave the way for hardware-in-the-loop testing and, eventually, integration of an LED-based payload into the STARI mission.

The following section will describe the methodology we aim to follow for the proposal. We will use existing STARI simulation datasets (obtained through the author's work in SLAB under Dr. Simone D'Amico) to simulate LED beacon formations in orbit, and then apply our position and pose estimation and filtering algorithms to compare the estimated state and the true state of the satellites. We will then discuss the results of our work and provide ideas to improve this system for the future.

II. Methodology

A. Coordinate System and Terminology Definition

We begin by defining our coordinate system, as shown in Figure 1. The Z axis (radial) points to the star, the X axis (along-track) points to the deputy satellite (known as the baseline vector between the satellites), and the Y (cross-track) axis closes the coordinate frame.

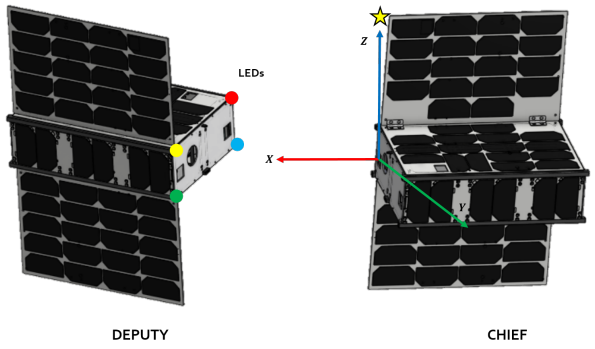


Fig. 1. Coordinate Frame Definition, including LEDs shown on the Deputy

We define f as the focal length of the camera, in meters. p_x and p_y are defined to be the physical pixel pitch in the x and y directions respectively, measured in meters/pixel. Then, $f_x = f/p_x$ and $f_y = f/p_y$ are the focal length, in pixels, along the x and y axes. Typically, p_x and p_y are equal, and we define a constant $M = f/p_x = f/p_y$. c_x and c_y represent the image center in pixels. s is defined to be the skew value, which is typically equal to zero unless the sensor has non-rectangular pixels.

We define the "state" of our satellite as a vector $\mathbf{x} \in \mathbb{R}^6$ composed of three positional components, measured in meters, and three orientational components, measured in radians.

$$\mathbf{x} = \begin{bmatrix} X & Y & Z & \theta & \phi & \psi \end{bmatrix}^T \tag{1}$$

θ represents a counter-clockwise positive rotation around the X-axis, colloquially referred to as the "roll", ϕ represents a counter-clockwise positive rotation around the Y-axis, colloquially referred to as the "pitch", and ψ represents a counter-clockwise positive rotation around the X-axis, colloquially referred to as the "yaw". The "optimal" state of the Deputy in the Chief's coordinate frame is then the state in which there are no angular rotations and no offsets in the Y and Z axes:

$$\mathbf{x}_{opt} = \begin{bmatrix} X & 0 & 0 & 0 & 0 & 0 \end{bmatrix}^T \tag{2}$$

For simplicity, **this project only provides an estimate of the relative position of the deputy with respect to the chief**. A methodology for full pose estimation is discussed in a later section. In addition, we ignore perturbations on the orbit, such as J2, drag and solar radiation pressure, to simplify the dynamics to be usable in a standard Kalman Filter. It is also worth noting that we're only considering position with no control input to correct for these in our model, i.e $\mathbf{u} = \mathbf{0}$.

We have knowledge of the following things in real-time:

- 4 known beacon LED positions on the Deputy, $\mathbf{X}_i \in \mathbb{R}^2$
- Corresponding detected image pixel positions, $\mathbf{x}_i \in \mathbb{R}^2$
- Camera intrinsic matrix, $\mathbf{K} \in \mathbb{R}^{3\times3}$

$$K = \begin{bmatrix} f_x & s & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \tag{3}$$

B. Obtaining Relative Position

We define d_{pix} to be the distance between two LEDs, in pixels, as seen by the camera, and d_{real} as the actual distance between LEDs on the cubesat, in meters. Then:

$$X = f_x \frac{d_{real}}{d_{pix}} \tag{4}$$

Once the along-track separation is found, the cross-track and radial separations can be found as follows:

$$Y = (u - c_x)\frac{X}{f_x}; \quad Z = (v - c_y)\frac{X}{f_y} \tag{5}$$

where (u, v) is the detected centroid in the image of the LED, in pixels.

C. Navigation Pipeline

The navigation system aboard STARI will likely consist of a multitude of sensor inputs to estimate the state vector. GPS is used to estimate coarse position, LEDs on the deputy are used to estimate fine relative attitude and position, and star trackers are used to estimate fine attitude. System dynamics will be used to estimate the system's position and attitude as well. This sensor triad, augmented with dynamics estimates, forms the core of the navigation stack, detailed in Figure 2. Measurements and estimated states are fed into the Kalman filter pipeline, to develop a state and associated estimate, which will be used in the attitude control system to maintain a constant relative orientation. Prior work has thoroughly demonstrated using GPS and star trackers for position and attitude estimation. This research aims to demonstrate the use of vision-based navigation with LEDs for relative position and attitude. For the scope of this project, only vision-based LED

navigation will be used in the sensor stream, and GPS and star-trackers are ignored for this iteration. The performance metric used to assess the performance of the Kalman Filter is the root mean squared error (RMSE) between the true data from the simulations and the filtered data.

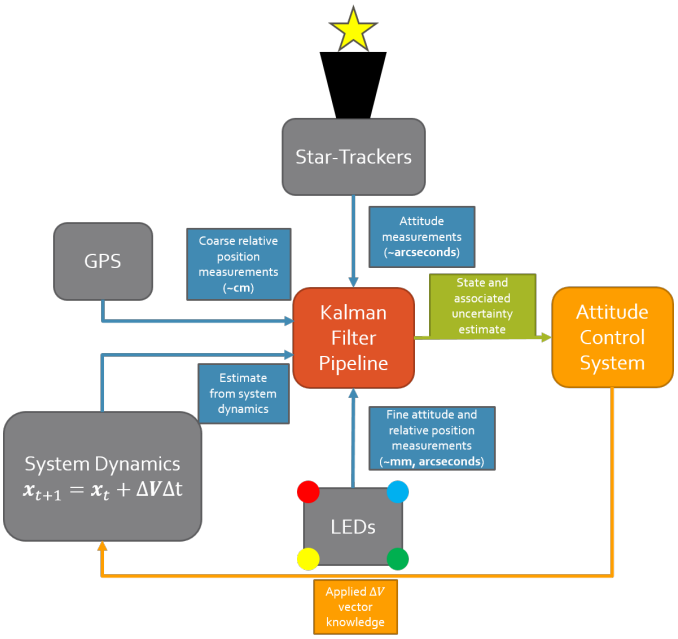


Fig. 2. Navigation System Overview

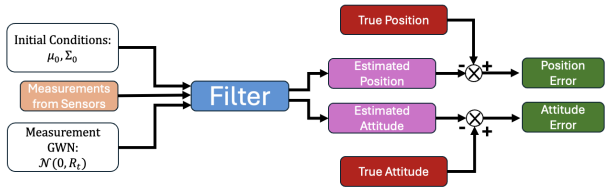


Fig. 3. Determining Filter Error

D. Dynamics Propagation

As part of the Kalman filter, we must compare our camera sensor measurements with our prediction of state through dynamics. For the scope of this project, only orbital dynamics is propagated, and attitude is not considered. There are various methods that can be used to propagate the dynamics of an orbit. These include a standard Keplerian propagator which propagates in terms of orbital elements, or a Fundamental Orbital Differential Equation Propagator, or FODE, which propagates in terms of position and velocity in the Earth-Centered Inertial (ECI) frame [12]. Because this is a distributed space systems proble, and what matters is the relative position between the chief and deputy, a relative motion propagator can be used. There are several options to choose from, including the Hill-Clohessy–Wiltshire (HCW) equations to propagate relative position and velocity [13], the Yamanaka-Ankerson equations propagate relative position and velocity in a non-linear form [14], and the ROE STM propagates relative orbital elements (ROEs) [15]. Because we represent state as relative position and velocity, an ROE propagator is not efficient, and in order to make the Jacobian calculations tractable, we choose to use the linear HCW relative motion propagator. This method has been thoroughly studied in literature, but has a simple caveat that it is only applicable for circular and near-circular orbits, which in our case is valid. The HCW state transition matrix (STM) is given by:

$$\mathbf{x}_{t+1} = \mathbf{A}\delta t \mathbf{x}_t + \mathbf{x}_t \tag{6}$$

Where $\mathbf{A}$ is:

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 3n^2 & 0 & 0 & 0 & 2n & 0 \\ 0 & 0 & 0 & -2n & 0 & 0 \\ 0 & 0 & -n^2 & 0 & 0 & 0 \end{bmatrix} \tag{7}$$

And $\mathbf{x}$ is the state vector:

$$\mathbf{x}_t = \begin{bmatrix} x \\ y \\ z \\ \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} \tag{8}$$

This method can be used to propagate the relative position between the chief and deputy, and will be used as the dynamics in our Kalman Filter.

E. Kalman Filter Pipeline

Although the Extended Kalman Filter (EKF) or Unscented Kalman Filter (UKF) are more common choices for spacecraft state estimation due to their higher precision, we selected the Kalman Filter (KF) for simplicity and due to the linearity of our propagation equations.

The ground-truth simulation data for our project is based on existing simulation work from the STARI mission. The measured LED positions in pixels are generated through the inverse of the pinhole camera model, with some simulated relative motions in the cross-track and radial directions. We transferred the pixel values back to the relative position using the pinhole camera model. Relative velocity is calculated using a central difference method on the position data. So, our measurement is programmed as a lookup table of relative position and velocity for each time step. For the "dirty" dataset (defined in a later section), due to the need for online outlier rejection, our measurement model first takes pixel values, filters them for outliers, then converts them to relative position and velocity data. This is probably the most realistic measurement model that can be implemented on a spacecraft.

$$\begin{aligned} \text{mean:} \quad \mu_{t|t-1} &= A_{t-1}\mu_{t-1} + B_{t-1}u_{t-1} \\ \text{cov:} \quad \Sigma_{t|t-1} &= A_{t-1}\Sigma_{t-1}A_{t-1}^T + Q_{t-1} \end{aligned}$$

$$\begin{aligned} \text{mean:} \quad \mu_{t|t} &= \mu_{t|t-1} + K_t(y_t - C_t\mu_{t|t-1}) \\ \text{cov:} \quad \Sigma_{t|t} &= \Sigma_{t|t-1} - K_t C_t \Sigma_{t|t-1} \\ \text{where} \quad K_t &= \Sigma_{t|t-1} C_t^T (C_t \Sigma_{t|t-1} C_t^T + R_t)^{-1} \end{aligned}$$

The default parameters used in the Kalman Filter are:

$$\mathbf{Q} = \text{diag}(10, 10, 10, 10, 10, 10) \tag{9}$$

$$\mathbf{R} = \text{diag}(1000, 1000, 1000, 1000, 1000, 1000) \tag{10}$$

Here μ is the estimated state. For the definition of other variables, we need to introduce the linear Gaussian state-space model.

$$\mathbf{x}_t = A_{t-1}\mathbf{x}_{t-1} + B_{t-1}\mathbf{u}_{t-1} + \mathbf{W}_{t-1}$$
$$\mathbf{y}_t = C_t\mathbf{x}_t + D_t\mathbf{u}_t + \mathbf{V}_t$$

where A_{t-1}, B_{t-1}, C_t, and D_t matrices are constants in the model. W_{t-1} and V_t are Gaussian White Noise (GWN) terms within the dynamics and measurement steps respectively.
For the dynamics step of the linear Gaussian state-space model. We are using linear HCW relative motion with no control input. Eq. 6 can be rewritten as:

$$\mathbf{x}_t = (\mathbf{A}\delta t + \mathbf{I}_{6\times 6})\mathbf{x}_{t-1} \tag{11}$$

The $\mathbf{A}_{t-1}$ and $\mathbf{B}_{t-1}$ matrices are defined as:

$$\begin{aligned} \mathbf{A}_{t-1} &= \mathbf{A}\delta t + \mathbf{I}_{6\times 6} \\ &= \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 3n^2 & 0 & 0 & 0 & 2n & 0 \\ 0 & 0 & 0 & -2n & 0 & 0 \\ 0 & 0 & -n^2 & 0 & 0 & 0 \end{bmatrix} \delta t + \mathbf{I}_{6\times 6}, \end{aligned} \tag{12}$$

where n is the mean motion of the chief spacecraft and δt is the time step size of the state-space model.

$$\mathbf{B}_{t-1} = \mathbf{0}_{6\times 3} \tag{13}$$

The GWN related to the dynamics step is defined as:

$$\mathbf{W}_{t-1} \sim \mathcal{N}(0, \mathbf{Q_{t-1}}) \tag{14}$$

The measurement is obtained from the lookup table at each time step, and the measurement model is a pseudo-measurement of the state itself. The measurement step can be expressed as:

$$\mathbf{y_t} = \mathbf{x_t} \tag{15}$$

The $\mathbf{C}_t$ and $\mathbf{D}_t$ matrices are defined as:

$$\mathbf{C}_t = \mathbf{I}_{6\times 6} \tag{16}$$

and

$$\mathbf{D}_t = \mathbf{0}_{6\times 3} \tag{17}$$

Similarly, the GWN related to the dynamics step is defined as:

$$\mathbf{W}_t \sim \mathcal{N}(0, \mathbf{Q_t}) \tag{18}$$

F. Data sets used

We ran our Kalman filter on two distinct data sets: One with injected Gaussian White Noise into the pixels, and one with GWN along with random starlight injected (henceforth referred to as the "dirty" dataset).

1) GWN data: To emulate the noise in images associated with CCDs used in cameras, we injected Gaussian White Noise into our original dataset with a covariance of 0.03, which is typical for CCDs based on [11]. This data is shown in Figure 4.

2) "Dirty" data: For this dataset, we injected GWN and starlight noise into our original data. We modeled the starlight as a Poisson process and, at each time step, randomly replaced between zero and three of the eight LED centroid pixel values in our dataset with a coordinate meant to emulate a star, implying that our vision-based system mistakenly interpreted a star as an LED. A future section will show our online filtering technique for starlight. Figure 5 shows the corruption of data using starlight noise, and Figure 6 shows the data after outlier rejection.

III. Results and Analysis

A. Data with Gaussian White Noise

For the dataset with only Gaussian White Noise, we observed that the estimated state converges to the ground truth very well, with a RMSE error of less than 1 meter for X-position and less than 2 millimeters for the Y and Z positions. The results can be seen in Figures 7 and 8.

B. "Dirty" Data Set

For the "dirty" dataset, we observed that the estimated state converges to the ground truth very well, albeit with random spikes corresponding to time-steps at which the pixels weren't adequately filtered for starlight noise. We observed a RMSE error of about 1 meter for X-position, less than 3 millimeters for the Y-position, and about 10 millimeters of error in the Z-position. The results can be seen in Figures 9, 10 and 11. It is worth noting that, instead of rejecting these stars, it is actually more realistic to account for these stars in the camera FOV by comparing them to star catalogs to obtain an added data stream for absolute attitude and position, akin to a star tracker.

IV. Conclusion

We have presented a novel LED-beacon based system for relative position estimation in space. Using LEDs and the properties of the camera, one can deduce the relative position (and orientation) of a spacecraft in a formation. We used a Kalman Filter to show that the error in the estimated position falls within the requirements of current missions that require tight relative positioning. We further investigated the case

where stars can be mistaken for LEDs, and presented an online outlier-rejection method to reject these occurrences. In this case, the errors in position were higher than the mission requirements. However, as previously discussed, the presence of stars in the FOV could actually be beneficial as an added data stream for absolute attitude, similar to a star tracker.

V. Future Work

The work in this paper can be expanded upon in the following ways:

- Incorporation of attitude dynamics for full pose estimation (methodology in the following section)
- Explore higher-fidelity orbital dynamics model
- Investigation of different outlier rejection methods
- Sensor fusion with simulated star-tracker and GPS data
- Stars in the LED camera FOV can be compared to star catalogs for added attitude knowledge
- Incorporation of a higher-fidelity orbital dynamics model
- Realistic estimation for Q and R
- Incorporation of a methodology to distinguish between LEDs, using either color or flickering frequencies, for unique solutions when solving for attitude

VI. Methodology for Full Pose Estimation

The following steps will be used to obtain relative position and attitude. This methodology of position and pose estimation was influenced by [16], [17], [18], and [19].

1) Build the Homography Matrix: The Homography matrix H can be found using the 2D-2D correspondence:

$$\mathbf{x}_i \sim \mathbf{X}_i \cdot H \tag{19}$$

H can be found using the function *findHomography* in Python's OpenCV library. H is then related to other parameters as follows:

$$H = K \begin{bmatrix} r_1 & r_2 & t \end{bmatrix} \tag{20}$$

Where K is the intrinsic matrix, r_1 and r_2 are the first and second columns of the rotation matrix R, and t is the translation vector.

2) Normalize Homography: We then remove the camera intrinsics from the homography matrix:

$$\hat{H} = K^{-1}H = \begin{bmatrix} \mathbf{h}_1 & \mathbf{h}_2 & \mathbf{h}_3 \end{bmatrix} \tag{21}$$

3) Recover Rotation Columns and Translation: From equation 21, the rotation and translation can be extracted through:

$$\begin{aligned} \lambda &= \frac{1}{\|\mathbf{h}_1\|} \\ \mathbf{r}_1 &= \lambda \cdot \mathbf{h}_1 \\ \mathbf{r}_2 &= \lambda \cdot \mathbf{h}_2 \\ \mathbf{r}_3 &= \mathbf{r}_1 \times \mathbf{r}_2 \\ \mathbf{t} &= \lambda \cdot \mathbf{h}_3 \end{aligned}$$

Then the rotation matrix is constructed as follows:

$$R = \begin{bmatrix} \mathbf{r}_1 & \mathbf{r}_2 & \mathbf{r}_3 \end{bmatrix} \tag{22}$$

This matrix can optionally be orthonormalized using SVD.

4) Extract Euler Angles: Finally, the attitude of the deputy spacecraft and the distance between the chief and the deputy can be obtained. Using a ZYX convention:

$$\phi = arcsin(-R_{3,1}) \tag{23}$$

$$\psi = atan2(R_{2,1}, R_{1,1}) \tag{24}$$

$$\theta = atan2(R_{3,2}, R_{3,3}) \tag{25}$$

5) Compute X, Y, Z: For real-world distance c_d and corresponding pixel distance c_{dp}, we can obtain the X distance as follows:

$$X = \frac{f_x \cdot c_d}{c_{dp} \cdot R_{3,3}} \tag{26}$$

Then, if (u, v) is the detected centroid in the image of the LED, in pixels Y and Z can be calculated:

$$Y = (u - c_x)\frac{X}{f_x}; \quad Z = (v - c_y)\frac{X}{f_y} \tag{27}$$

VII. Code

The code for the project can be found at: https://code.stanford.edu/prachetj/aa273-final-project/-/tree/old-state?ref_type=heads.
The working branch is labeled old-state.

Appendix A
Additional Results for "Dirty" Data Set

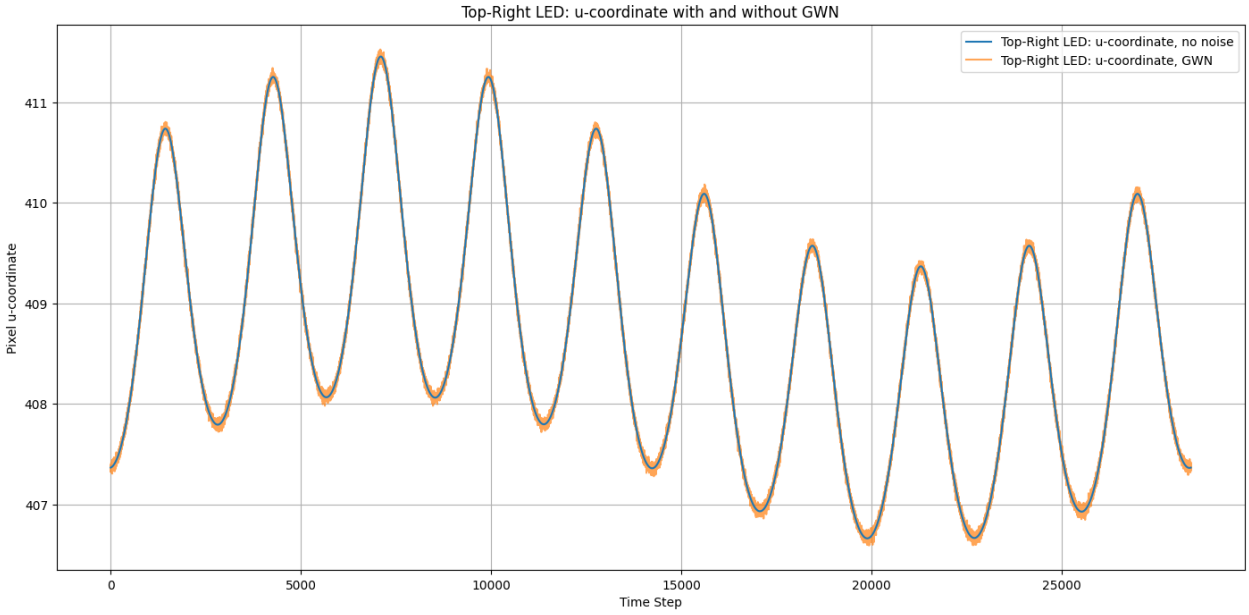


Fig. 4. Comparison of clean data and data with GWN

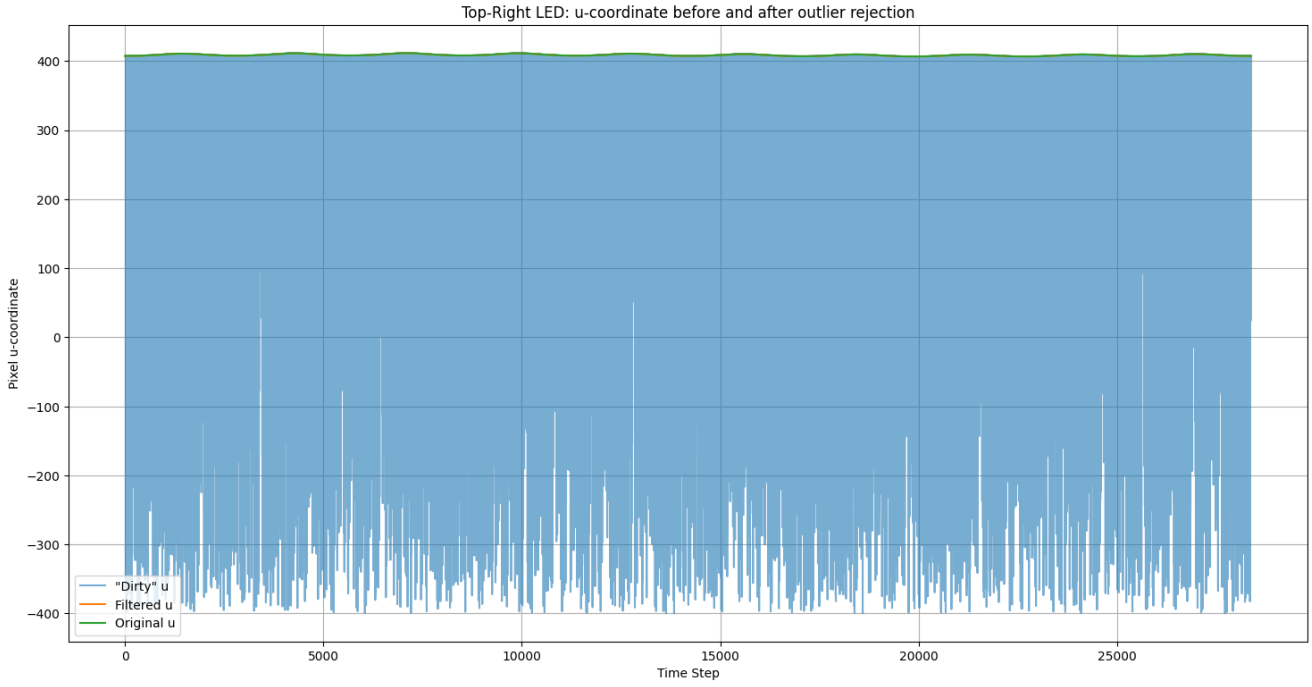


Fig. 5. “Dirty” Data Set

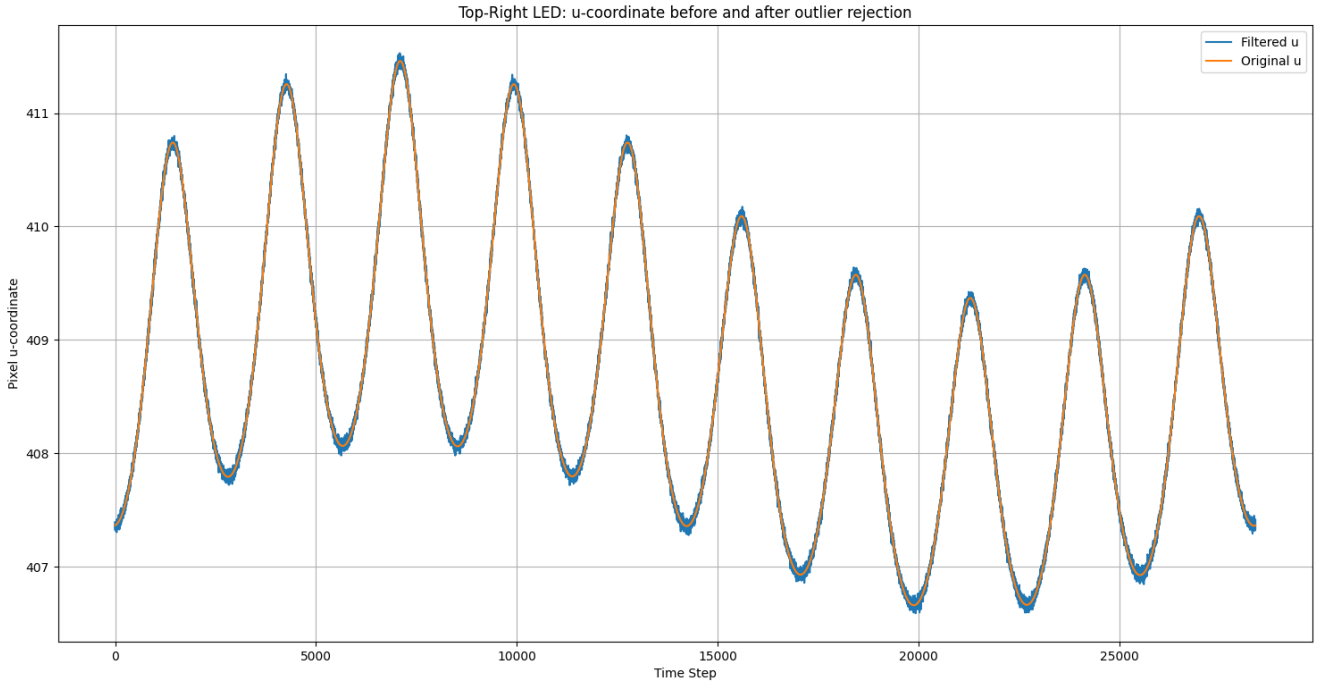


Fig. 6. "Dirty" Data Set after Outlier Rejection

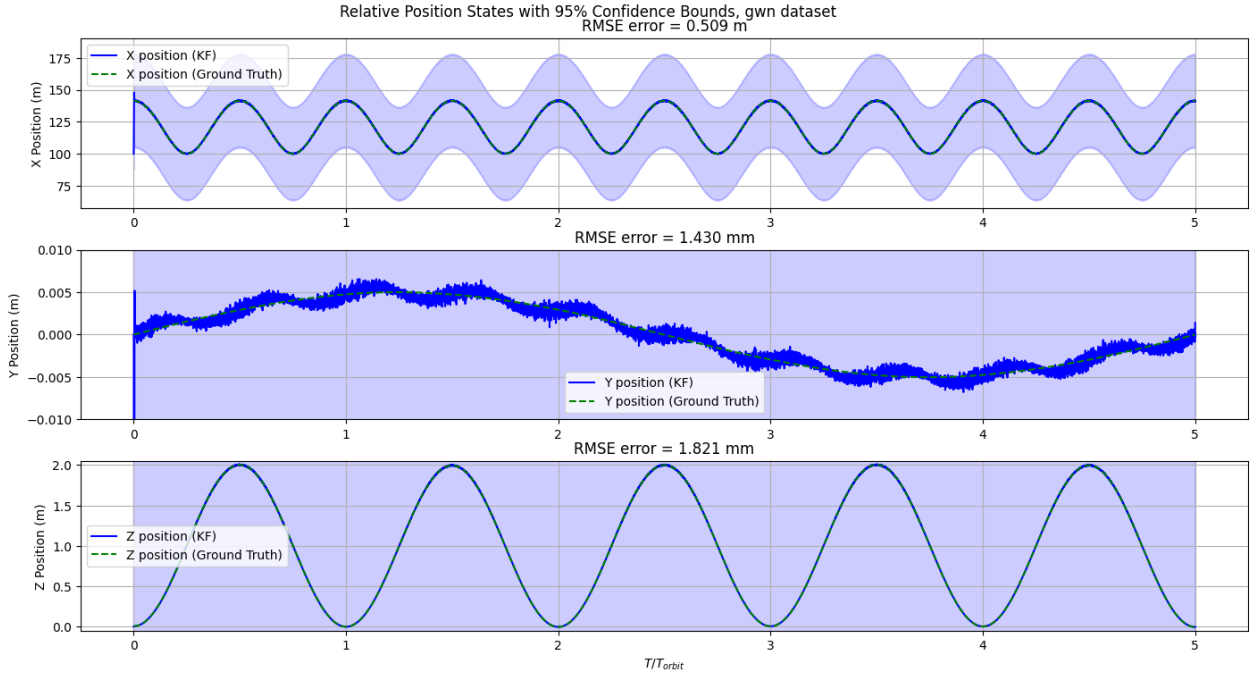


Fig. 7. GWN Data Set after Kalman Filtering

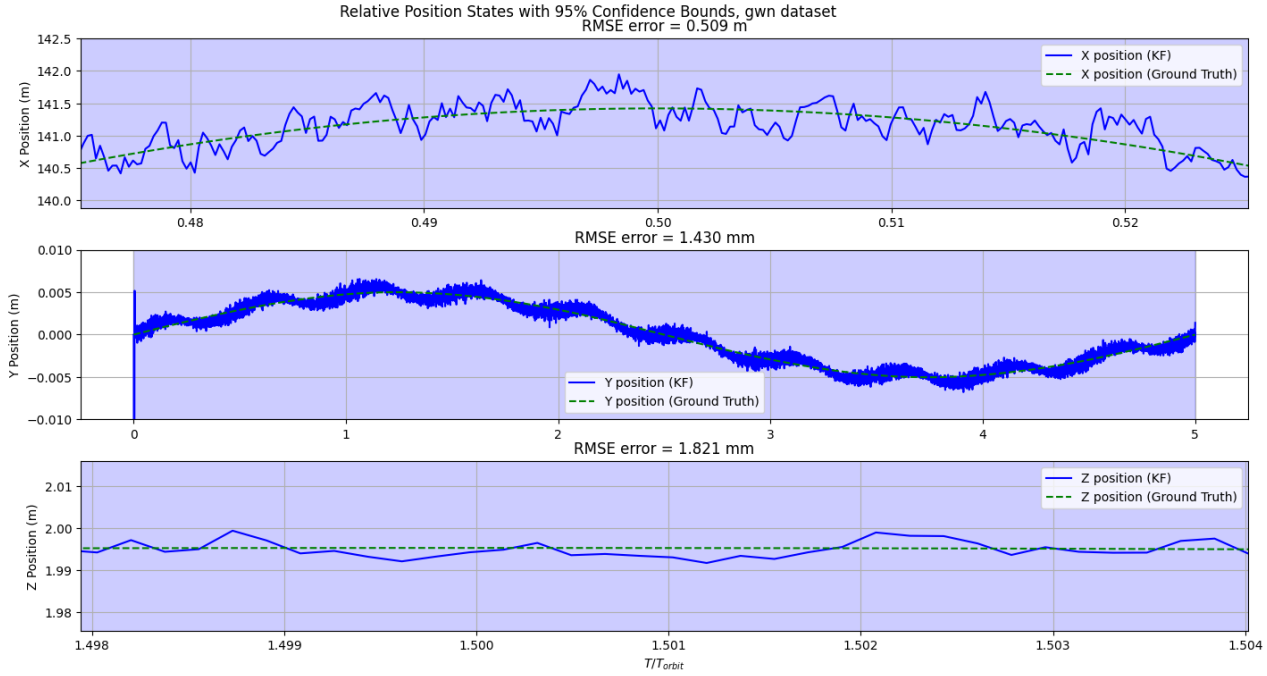


Fig. 8. GWN Data Set after Kalman Filtering, zoomed

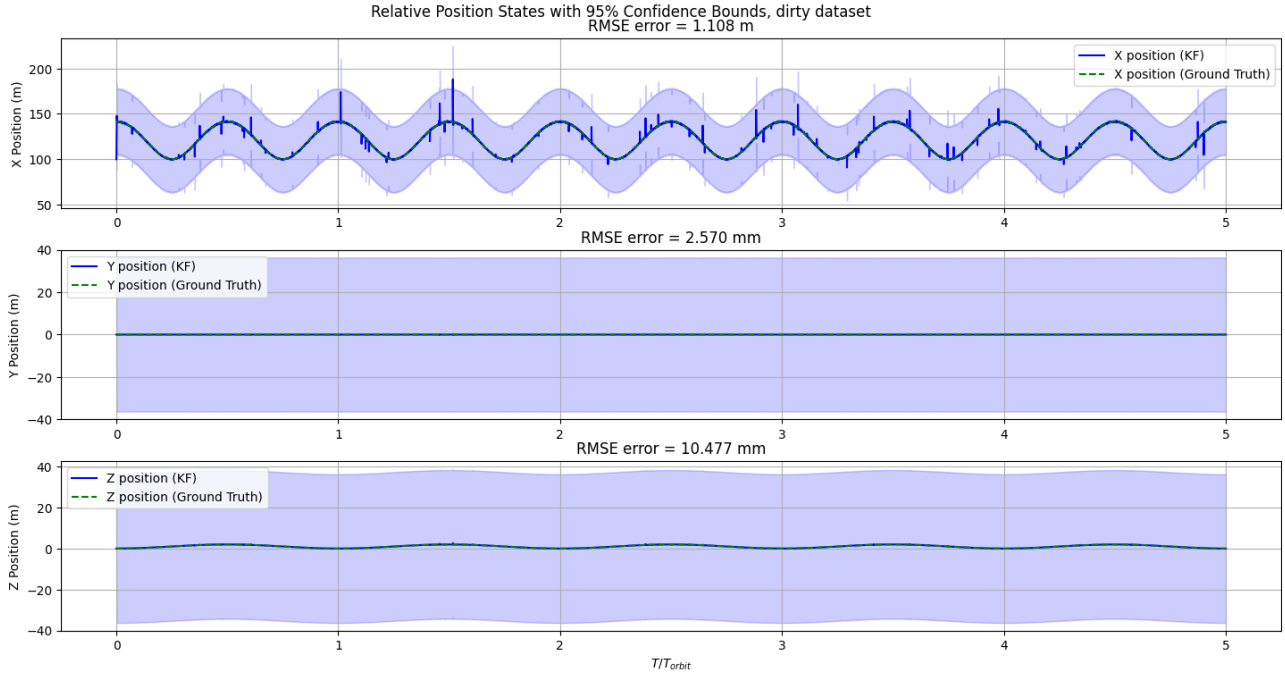


Fig. 9. Dirty Data Set after Kalman Filtering

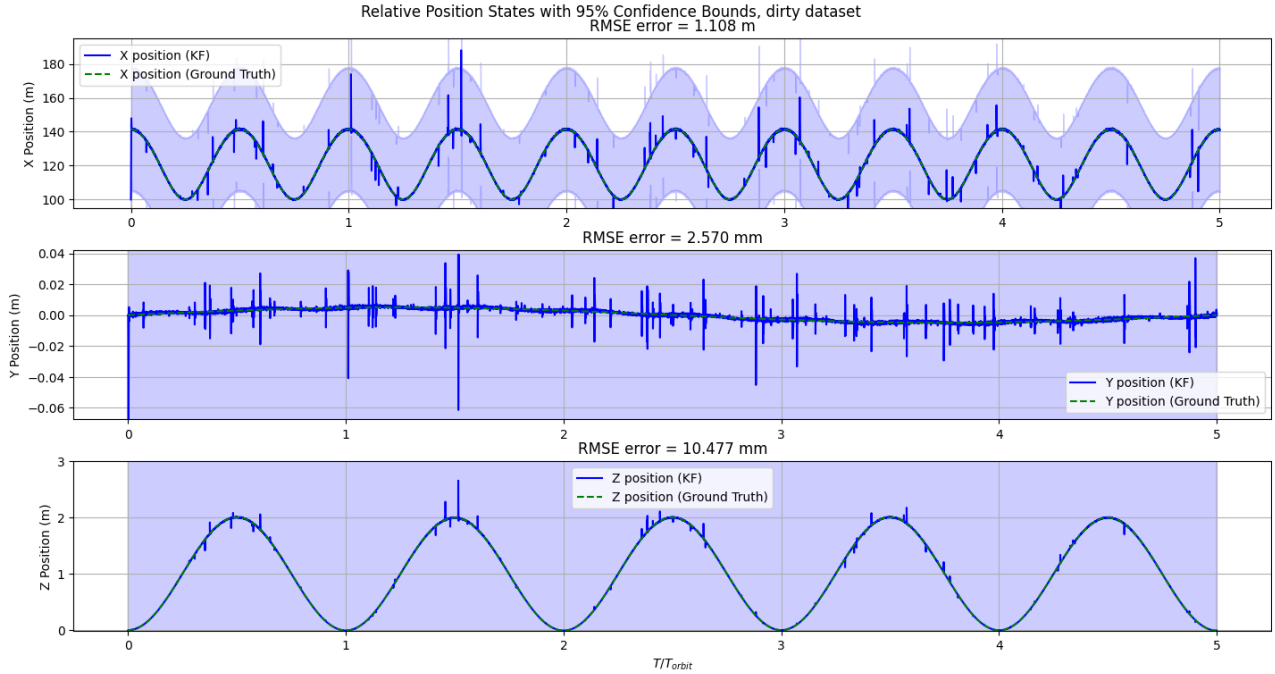


Fig. 10. Dirty Data Set after Kalman Filtering, zoomed

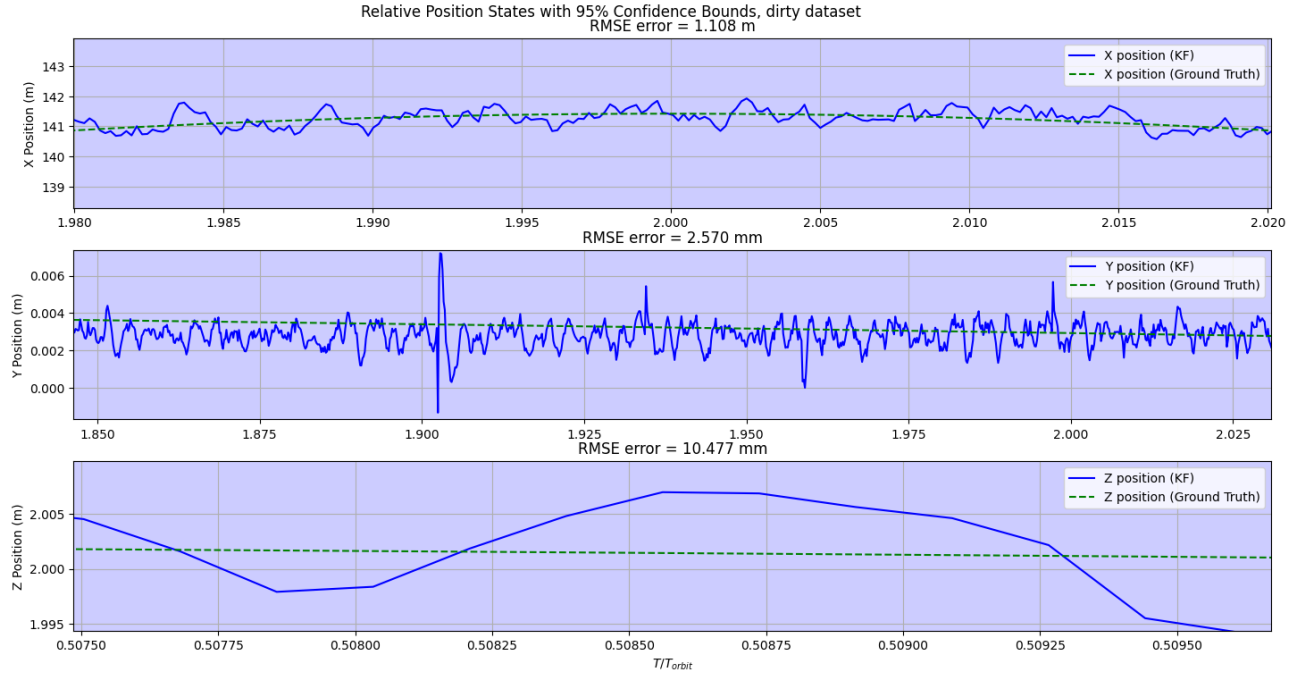


Fig. 11. Dirty Data Set after Kalman Filtering, zoomed for further detail

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